

1) Underline one of the selections either T or F.

1)	Cobol language has dominated business domain for a long time	T	F
2)	Fortran language has dominated artificial intelligence for a long time	T	F
3)	The orthogonality sounds in both reliability and wirtability	T	F
4)	In logic programming language, such as prolog, rules are specified in a particular order	T	F
5)	A derivation is a repeated application of rules, starting with a non-terminal symbol and ending with terminal symbols	T	F
6)	The compilers are used to translate machine code into source code	T	F
7)	In the left hand side, in BNF rules, consists of terminal and non-terminal symbols	T	F
8)	A context-free grammar is said to be ambiguous if and only if some string in the language has one parse tree	T	F
9)	The acronym BNF stands for Backward-Normal Form	T	F
10)	The operator in an arithmetic expression which is generated lower in the parse tree has lower precedence than an operator that produced higher up in the tree	T	F

2. Compute the weakest precondition for each of the following statements and postconditions (show your answer):

A. $y = 3 * x + 6 \{x < 0\}$

B. $a = 5 * a + 6$
 $a = a + 3 \{ a < 0 \}$

First Quiz, CP

Model: A

Name: -----

1) Underli

1)	A conte
2)	language
3)	Cobol la
	Prolog la
	TL

3. Prove the following grammar is ambiguous:

$$\langle X \rangle \rightarrow \langle Y \rangle$$

$$\langle Y \rangle \rightarrow \langle Y \rangle + \langle Y \rangle / \langle W \rangle$$

$$\langle W \rangle \rightarrow x / y / z$$

First Quiz, CPCS301

Model: A

Name: ----- ID:----- Section:-----

1) Underline one of the selections either T or F.

1)	A context-free grammar is said to be ambiguous if and only if some string in the language has more than one parse tree	T	F
2)	Cobol language has dominated scientific computing for a long time	T	F
3)	Prolog language has dominated artificial intelligence for a long time	T	F
4)	The acronym BNF stands for Backus-Naur Form	T	F
5)	The orthogonality sounds in both readability and wirtability	T	F
6)	In logic programming language, such as prolog, rules are specified in a particular order	T	F
7)	The compilers are used to translate source code into machine code	T	F
8)	In the left hand side, in BNF rules, consists of terminal and non-terminal symbols	T	F
9)	A derivation is a repeated application of rules, starting with a terminal symbol and ending with non-terminal symbols	T	F
10)	The operator in an arithmetic expression which is generated lower in the parse tree has precedence over an operator that produced higher up in the tree	T	F

2. Compute the weakest precondition for each of the following statements and postconditions (show your answer):

A. $x = 3 * y + 6 \{x > 0\}$

B. $x = 3 * x + 6$
 $x = x + 3 \{x < 10\}$

2. Prove the following grammar is ambiguous:

$$\langle L \rangle \rightarrow \langle T \rangle$$

$$\langle T \rangle \rightarrow \langle T \rangle * \langle T \rangle \mid \langle id \rangle$$

$$\langle id \rangle \rightarrow a \mid b \mid c$$